

1D-CNN SYSTEM ARCHITECTURE PIPELINE

[01] High-Fidelity Signal Capture

Captures 3-phase signals (Va, Vb, Vc, Ia, Ib, Ic) at 10kHz sampling. Data is converted to Per-Unit (p.u.) values to ensure the model is invariant to base voltage levels across the network.

TECHNICAL SPECIFICATIONS:

Sampling: 10,000 Hz

Channels: 7 (Time + 6 Signals)

Domain: Per-Unit Normalized

[02] Temporal Windowing

Segments the continuous stream into 200 time-step windows (20ms), corresponding to one fundamental 50Hz cycle. Overlapping strides augment the dataset and capture transient starts.

TECHNICAL SPECIFICATIONS:

Window Size: 200 Samples

Stride: 50 Samples

Buffer Duration: 20ms

[03] Neural Preprocessing

Standard normalization (z-score) removes DC offsets and scales features to zero mean and unit variance, ensuring numerical stability during backpropagation.

TECHNICAL SPECIFICATIONS:

Method: StandardScaler

Artifact: scaler.joblib

Formula: $z = (x - \text{mean}) / \text{std}$

[04] Deep CNN Feature Extractor

Utilizes three convolutional blocks (64, 128, 256 filters) with Batch Normalization and ReLU. It learns complex temporal dependencies and phase-shift correlations from the multi-channel tensor.

TECHNICAL SPECIFICATIONS:

Layers: 3x [Conv1D + BN + ReLU + MaxPool]

Kernels: 7, 5, 3

Pooling: GlobalAveragePooling1D

[05] Multi-Task Heads

The architecture splits into three parallel heads: Sigmoid for binary fault status, Softmax for scenario type, and a scaled Sigmoid for physical distance estimation (0-5 KM).

TECHNICAL SPECIFICATIONS:

Heads: 1x Sigmoid (Status), 1x Softmax (Type), 1x Regressor (Loc)

Loss Weights: 1.0, 1.0, 0.5

[06] Diagnostic Resolution

Converts AI outputs into engineering units. Distance is computed via a scaling factor and signal magnitudes are mapped to kV/kA for end-user interpretation.

TECHNICAL SPECIFICATIONS:

Distance: Pred * 5.0 KM

Voltage Base: 200 kV

Current Base: 1 kA